

Levinson's method in short intervals and simple zeros of the zeta function

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CAOS Research program

github.com/fsantibanezleal/CAOS_RESEARCH

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Abstract

For $1/2 < \theta < 1$, we prove a uniformly shifted mollified second moment on windows of length T^θ for mollifier exponents $\nu < \min\{1/2, 2\theta - 1, (5\theta - 2)/6\}$. An exact composite additive-divisor transformation gives a Mellin integral whose arithmetic dependence is through $\log(pq/n)$. Grouping equal rational frequencies and retaining their multiplicities permits elementary mean-square estimates, proved both with a Gaussian Schur bound and with direct interval integration. Weighted Cauchy–Schwarz then gives the relative errors TM/H^2 and $TM^3/H^{5/2}$. Complex shifts, both residues, unbalanced blocks, infinite dual tails and smooth-window conversion are retained. An alternative trilinear estimate gives the complementary range $\nu < \min\{1/2, (17/33)(2\theta - 1)\}$. The moment supports Conrey’s operator polynomial of arbitrary fixed degree. Localized counting and Wang’s pair theorem with a finite parity inequality yield positive simple-critical density for every fixed $\theta \in [0.527, 1)$; an unchanged degree-201 detector gives lower density greater than 0.0005947542001 at $\theta = 0.527$. The estimates are asymptotic, give no effective starting height, use attributed analytic inputs including Wang’s recent preprint, and do not prove the Riemann hypothesis.

1 Introduction

Let $\rho = \beta + i\gamma$ denote the nontrivial zeros of the Riemann zeta function. For $T, H > 0$ let $N(T, H)$ count the zeros with $T < \gamma \leq T + H$, with multiplicity; let $S(T, H)$ count the simple zeros among them with $\beta = 1/2$; and let $O(T, H)$ count the distinct points $t \in (T, T + H]$ at which Hardy’s function $Z(t) = e^{i\vartheta(t)}\zeta(1/2 + it)$ changes sign, that is, the distinct critical zeros of odd multiplicity. Throughout, $H = T^\theta$ with a fixed $\theta \in (1/2, 1)$ and $L = \log T$.

Selberg proved that $O(T, T^\theta) \gg N(T, T^\theta)$ for every $\theta > 1/2$, and Karatsuba extended this to $\theta > 27/82$ [7]. These theorems do not give simple zeros. For simple critical zeros, Steuding localized Levinson’s method with a linear operator polynomial, proved a short-window mollified

moment with error $O(T^{1/3+\epsilon}M^{4/3})$, and obtained a positive proportion for $\theta \geq 0.552$ [12, 13]; he also observed that the mollifier range available from the classical off-diagonal treatment gives only $\theta \geq 0.693$ for linear Q . Wang's short-interval pair-correlation theorem [14] gives the lower density

$$c(\theta) = 2 - \frac{\theta}{2} - \frac{1}{\sqrt{2}} \cot \frac{\theta}{\sqrt{2}}, \quad (1)$$

which is positive for $\theta > \theta_0 = 0.550193964744\dots$. The finite Hilbert–parity inequality of [4, Proposition 2.5] converts any lower density k for O/N into the simple-critical density

$$h(\theta; k) = \frac{3 + k - \sqrt{(1-k)(9-k-8c(\theta))}}{4}. \quad (2)$$

With the rank-six Selberg detector of Pearce-Crump [8], $k_6(\theta) = (\theta - 1/2)/(4eC_6)$ with $C_6 = 0.65663\dots$, the resulting term is positive exactly for $\theta > \theta_6$, where $0.5458837 < \theta_6 < 0.5458838$ [4, Theorem 1.2]. Since the positivity condition is $k > 1 - 2/(2 - c(\theta))$, the onset is governed by the size of the odd-support density near $\theta = 1/2$, where k_6 has slope only $1/(4eC_6) \approx 0.140$.

The localized Levinson detector supplies the odd-support density needed by the parity inequality. The elementary diagonal range is $\nu < \theta - 1/2$. Two estimates of the complete signed off-diagonal give

$$\nu_A(\theta) = \min\{1/2, 2\theta - 1, (5\theta - 2)/6\}, \quad (3)$$

$$\nu_B(\theta) = \min\{1/2, (17/33)(2\theta - 1)\}, \quad (4)$$

$$0 < \nu < \nu_*(\theta) := \max\{\nu_A(\theta), \nu_B(\theta)\}. \quad (5)$$

The first uses elementary rational-frequency mean-square bounds. The second uses the attributed trilinear theorem of Bettin and Chandee [1]. The first is stronger below $\theta = 12/13$, and the second is stronger above it. Within ν_A , the two error restrictions cross at $\theta = 4/7$. The labels [D] and [MV] mark mathematical deductions and finite machine-verified assertions, respectively.

Theorem 1.1 (Simple critical zeros from $T^{0.527}$, D+MV). *For every fixed $\theta \in [0.527, 1)$,*

$$\liminf_{T \rightarrow \infty} \frac{S(T, T^\theta)}{N(T, T^\theta)} \geq \max \left\{ 0, c(\theta), \frac{c(\theta) + 2\kappa_\theta}{3}, h(\theta; \kappa_\theta) \right\} > 0, \quad (6)$$

where κ_θ can be any certified detector in Table 1 whose ν satisfies (5). In particular the fixed detector with $\nu = 499/10000$ has $\kappa \geq 71589990975836346694543352023356087639/(2 \cdot 10^{39})$ and

$$h(527/1000; \kappa) > 0.0005947542001.$$

The same fixed detector is admissible for every larger $\theta < 1$.

The improvement in range is analytical; no new detector optimization is needed for the displayed consequence. The moment and counting statements are as follows.

Theorem 1.2 (Mollified moment on a short window, D). *Let ν satisfy (5), let $R > 0$, and let P be a real polynomial with $P(0) = 0$, $P(1) = 1$, and let Q be a real polynomial with $Q(0) = 1$. Put $\sigma_0 = 1/2 - R/L$, $y = T^\nu$, $\Delta = H/L$,*

$$V(s) = Q\left(-\frac{1}{L} \frac{d}{ds}\right) \zeta(s), \quad \psi(s) = \sum_{h \leq y} \frac{\mu(h)}{h^{s+1/2-\sigma_0}} P\left(\frac{\log(y/h)}{\log y}\right).$$

Let w be smooth with $0 \leq w \leq 1$, $w = 1$ on $[T, T+H]$, support in $[T-\Delta, T+H+\Delta]$ and $w^{(j)} \ll_j \Delta^{-j}$. Then

$$\int_{\mathbb{R}} w(t) |V\psi(\sigma_0 + it)|^2 dt = c(P, Q, R, \nu) \hat{w}(0) + O(H/L),$$

$$c(P, Q, R, \nu) = 1 + \frac{1}{\nu} \int_0^1 \int_0^1 (w_R(v) P'(u) + \nu w'_R(v) P(u))^2 du dv, \quad w_R(v) = e^{Rv} Q(v).$$

Theorem 1.3 (Distinct sign changes, D). *In the setting of Theorem 1.2, suppose that $Q(x) + Q(1-x)$ is a nonzero constant. Then*

$$\liminf_{T \rightarrow \infty} \frac{O(T, T^\theta)}{N(T, T^\theta)} \geq \kappa(P, Q, R, \nu) := 1 - \frac{1}{R} \log c(P, Q, R, \nu).$$

The constant is Conrey's [6] in Young's form [15]. The counting argument pays full weight for detector zeros on the critical line; a polynomial of higher degree supplies distinct odd-order zeros, which are the appropriate input for the parity inequality. Conrey's Gaussian-window proposition applies under a different restriction, $1 - \theta < 1/7 - \nu/4$. Bounded primary-source searches did not locate the shifted short-window range (5) or the onset 0.527; this does not establish worldwide priority. The global 17/33 range is context [1], not a short-window theorem imported without proof.

2 The elementary diagonal range

Throughout this section assume $0 < \nu < \theta - 1/2$. It supplies the arithmetic main-term evaluation and localized counting inputs; the signed estimate of Section 3 proves the full range (5).

We follow Young [15]. His Theorem 2 treats a weight supported in $[T/4, 2T]$ with $w^{(j)} \ll (T/L)^{-j}$ and a mollifier of length $M = T^\vartheta$, $\vartheta < 1/2$. Here the weight has support of length at most $2H$, the derivative scale is $\Delta = H/L$, and M is replaced by $y = T^\nu$. We list every step in which the length enters; all others are used verbatim. As in Young's Section 3, it suffices to prove

$$I(\alpha, \beta) := \int w(t) \zeta(\tfrac{1}{2} + \alpha + it) \zeta(\tfrac{1}{2} + \beta - it) |\psi(\sigma_0 + it)|^2 dt = c(\alpha, \beta) \hat{w}(0) + O(H/L) \quad (7)$$

uniformly for $\alpha, \beta \ll 1/L$, with $c(\alpha, \beta)$ Young's (3.3) with ϑ replaced by ν and M by y . We first work on the annuli $\alpha, \beta \asymp 1/L$, $|\alpha + \beta| \gg 1/L$ of Young's Lemma 6. There the factor $G(s) = e^{s^2} p(s)$, $p(s) = ((\alpha + \beta)^2 - 4s^2)/(\alpha + \beta)^2$, of Young's approximate functional equation (Lemma 4) satisfies $p(s) \ll L^2(1 + |s|^2)$. Hence the weight $V_{\alpha, \beta}(x, t)$ and the factor $X_{\alpha, \beta, t}$ of that lemma obey, for $t \geq T/2$,

$$t^j \partial_t^j V_{\alpha, \beta}(x, t) \ll_{A, j} L^2 (1 + x/t)^{-A}, \quad X_{\alpha, \beta, t} = (t/2\pi)^{-\alpha - \beta} (1 + O(t^{-1})). \quad (8)$$

The first bound is Young's (4.3), whose decay factor is printed as $(1 + |t/x|)^{-A}$; the Mellin representation (4.1) gives $(1 + x/t)^{-A}$, which is the form used in his proof of Lemma 5. The Stirling expansion (4.2) of $g_{\alpha, \beta}(s, t)$ is uniform only for $|s| = o(t^{1/2})$; on every vertical line used below, the part $|\Im s| \geq T^{1/3}$ is negligible because $|G(s)| \ll L^2 e^{-(\Im s)^2/2}$ there, while $|g_{\alpha, \beta}(s, t)| \ll (t + |s|)^{O(1)}$.

Lemma 2.1 (Twisted integral). *Let $h, k \leq y$ and let α, β lie on the annuli. Then*

$$\begin{aligned} \int w(t) \left(\frac{h}{k}\right)^{-it} \zeta(\tfrac{1}{2} + \alpha + it) \zeta(\tfrac{1}{2} + \beta - it) dt &= \sum_{hm=kn} \frac{\int V_{\alpha, \beta}(mn, t) w(t) dt}{m^{1/2 + \alpha} n^{1/2 + \beta}} \\ &+ \sum_{hm=kn} \frac{\int V_{-\beta, -\alpha}(mn, t) X_{\alpha, \beta, t} w(t) dt}{m^{1/2 - \beta} n^{1/2 - \alpha}} + O_A(T^{-A}). \end{aligned}$$

Proof. Insert the approximate functional equation. Its error term contributes $O(HT^{-A})$ and the terms with $hm = kn$ are the main terms. For $hm \neq kn$, the bounds (8) and $w^{(j)} \ll \Delta^{-j}$ give $\partial_t^j [w(t)V_{\alpha,\beta}(x,t)] \ll L^2(1+x/T)^{-A}\Delta^{-j}$ on the support, so j integrations by parts give

$$\int w(t) \left(\frac{hm}{kn}\right)^{-it} V_{\alpha,\beta}(mn,t) dt \ll HL^2 \frac{(1+mn/T)^{-A}}{(\Delta|\log(hm/kn)|)^j} \leq HL^2(1+mn/T)^{-A} \left(\frac{2\sqrt{hkmn}}{\Delta}\right)^j,$$

since $|\log(hm/kn)| \geq 1/(2\sqrt{hkmn})$ for $hm \neq kn$. Put $\eta = \theta - 1/2 - \nu > 0$ and $\epsilon = \eta/2$. If $mn \leq T^{1+\epsilon}$ then $2\sqrt{hkmn}/\Delta \leq 2yT^{(1+\epsilon)/2}L/T^\theta = 2LT^{-3\eta/4}$, and there are at most T^2 such pairs (m,n) , so these terms contribute $O(T^{-A})$ once $j > 4(A+5)/(3\eta)$. If $mn > T^{1+\epsilon}$ we do not integrate by parts: the trivial bound $HL^2(mn/T)^{-A'}$ summed against $(mn)^{-1/2+O(1/L)}$ gives $O(HL^3T^{(1+\epsilon)/2-\epsilon A'+o(1)})$, which is $O(T^{-A})$ for $A' \geq (A+3)/\epsilon$. The reflected part is identical because $t^j \partial_t^j X_{\alpha,\beta,t} \ll |X_{\alpha,\beta,t}| \ll 1$. $\square$

Inserting the mollifier, $I = I_1 + I_2 + O(T^{-A})$, where I_1 is Young's (5.1) with M replaced by y and I_2 is $I_1(-\beta, -\alpha)$ with $X_{\alpha,\beta,t}$ inserted in the t -integral. On the support of w we have $t = T(1 + O(H/T))$, so by (8) $X_{\alpha,\beta,t} = (T/2\pi)^{-\alpha-\beta}(1 + \rho(t))$ with $\rho \ll H/T$. Write $h = gh'$, $k = gk'$ with $(h', k') = 1$; then $hm = kn$ means $m = k'r$, $n = h'r$, and

$$\sum_{h,k \leq y} \frac{1}{\sqrt{hk}} \sum_{hm=kn} \frac{(1+mn/T)^{-A}}{(mn)^{1/2-O(1/L)}} \ll L \sum_{g \leq y} \frac{1}{g} \left(\sum_{h' \leq y} \frac{1}{h'} \right)^2 \ll L^4.$$

Bounding with absolute values, the contribution of ρ to I_2 is $\ll H(H/T)L^6 = o(H/L)$ because $\theta < 1$. Since $(2\pi)^{\alpha+\beta} = 1 + O(1/L)$ and $|I_1(-\beta, -\alpha)| \ll H$ by the next lemma (the region of the annuli is invariant under $(\alpha, \beta) \mapsto (-\beta, -\alpha)$),

$$I(\alpha, \beta) = I_1(\alpha, \beta) + T^{-\alpha-\beta} I_1(-\beta, -\alpha) + O(H/L). \quad (9)$$

Lemma 2.2 (Diagonal). *Uniformly on the annuli, $I_1(\alpha, \beta) = c_1(\alpha, \beta)\hat{w}(0) + O(H/L)$ with*

$$c_1(\alpha, \beta) = \frac{1}{(\alpha + \beta) \log y} \int_0^1 (P'(u) + \alpha \log y P(u))(P'(u) + \beta \log y P(u)) du.$$

Proof. Follow Young's Section 6. After the Mellin representation (6.1) and the arithmetic identity (6.2), move the u, v -contours to $\Re = \delta$ and the s -contour to $\Re s = -\delta + \epsilon_1$ with a fixed small $\epsilon_1 > 0$. The only pole crossed is $s = 0$, because G vanishes at the pole $s = -(\alpha + \beta)/2$ of $\zeta(1 + \alpha + \beta + 2s)$. The only factor involving w is $\int w(t)g_{\alpha,\beta}(s,t)dt$, and by Young's (4.2), $|g_{\alpha,\beta}(s,t)| \ll (t/2\pi)^{-\delta+\epsilon_1}(1+|s|^2)$ on the new contour for $|\Im s| \leq T^{1/3}$. This factor is therefore $\ll HT^{-\delta+\epsilon_1}(1+|s|^2)$, while $|y^{u+v}| = y^{2\delta}$; the remaining factors are as in Young. The new contour contributes $\ll HT^{-\delta(1-2\nu)+\epsilon_1}L^{O(1)} = o(H/L)$ for $\epsilon_1 < \delta(1-2\nu)/2$, since $\nu < 1/2$. The residue at $s = 0$ has $g_{\alpha,\beta}(0,t) = G(0) = 1$ and equals $\hat{w}(0)$ times the expression in Young's (6.3). Young's Lemma 7 and Section 7 involve neither w nor T and give the main term with relative error $O(1/L)$. $\square$

Proof of Theorem 1.2. As in Young's deduction of his Lemma 3 from Lemma 6,

$$c_1(\alpha, \beta) + c_1(-\beta, -\alpha) = \int_0^1 2PP' = 1, \quad \frac{1 - T^{-\alpha-\beta}}{(\alpha + \beta) \log y} = \frac{1}{\nu} \int_0^1 T^{-v(\alpha+\beta)} dv.$$

With (9) and Lemma 2.2 this gives (7) on the annuli, and the maximum modulus principle extends it to $\alpha, \beta \ll 1/L$ because both sides are holomorphic. By Young's (3.4), the left side of Theorem 1.2 equals

$$Q\left(-\frac{1}{L}\frac{\partial}{\partial\alpha}\right)Q\left(-\frac{1}{L}\frac{\partial}{\partial\beta}\right)I(\alpha, \beta)\Big|_{\alpha=\beta=-R/L};$$

the derivatives are Cauchy integrals over circles of radius $\asymp 1/L$ on which the error is uniform, and the operator applied to $c(\alpha, \beta)$ gives Young's (1.3) with ϑ replaced by ν . Since

$$\frac{d}{dx}\left[e^{R\nu x}P(x+u)Q(v+\nu x)\right]_{x=0} = e^{-Rv}(w_R(v)P'(u) + \nu w'_R(v)P(u)),$$

this is the stated constant. $\square$

Remark 2.3. The hypothesis $\nu < \theta - 1/2$ is used only in Lemma 2.1, and the bound $\nu < 1/2$ only in Lemma 2.2. The hypothesis $\theta < 1$, through $H = o(T)$, places the support of w in $[T/2, 2T]$, so that $t^{-1} \leq \Delta^{-1}$, and bounds ρ . The normalization $Q(0) = 1$ enters only through the constant term: the operator maps the term 1 of $c(\alpha, \beta)$ to $Q(0)^2$, so for every real polynomial Q the moment is $(Q(0)^2 + \nu^{-1} \int_0^1 \int_0^1 (w_R P' + \nu w'_R P)^2) \widehat{w}(0) + O(H/L)$, and in particular $O(H)$.

3 The signed estimate for a longer mollifier

All orders in this section are fixed before T tends to infinity. Write $e(x) = e^{2\pi i x}$ and $\sigma_{a,b}(n) = \sum_{uv=n} u^{-a} v^{-b}$. Shifts lie in a fixed $C/\log T$ disk, enlarged by a fixed factor when taking derivatives.

3.1 An exact Gaussian reduction and its summable remainder

For the Gaussian of width H , define

$$K_\beta(x) = \frac{2x^{-\beta}}{\sqrt{\pi}} \int_{\mathbb{R}} e^{-y^2 + (1-2\beta+2iT)y/H} \cos(2\pi x e^{2y/H}) dy.$$

Functional equation and Mellin inversion give

$$\begin{aligned} & \int_{\mathbb{R}} \zeta\left(\frac{1}{2} + \alpha + it\right) \zeta\left(\frac{1}{2} + \beta - it\right) (h/k)^{it} e^{-(t-T)^2/H^2} dt \\ &= 2\pi \sqrt{k/h} \sum_{n \geq 1} \sigma_{\alpha, -\beta}(n) K_\beta(nk/h) - 2\pi \zeta(\alpha + \beta) G(1 - \alpha) (h/k)^{1/2 - \alpha}, \end{aligned} \quad (10)$$

where $G(s) = \exp((s - 1/2 - iT)^2/H^2)$. For justification substitute $u = e^{2y/H}$ in the kernel and regularize its x Mellin transform by $e^{-\epsilon x}$. Dominated convergence and Gaussian integration give

$$\widetilde{K}_\beta(s) = 2(2\pi)^{-(s-\beta)} \Gamma(s - \beta) \cos(\pi(s - \beta)/2) G(s).$$

Move the moment's s contour from $1/2$ to the absolute Dirichlet half-plane. The pole $s = 1 - \alpha$ is crossed with the displayed minus sign. The apparent pole $1 + \beta$ is canceled by the cosine zero. The kernel has arbitrary inverse-power decay at infinity; its defining log-Gaussian amplitude and all derivatives have integrable endpoints.

On a fixed central band $x \asymp T$, expand only the nonlinear phase $\exp[\epsilon 2\pi i x(e^{2y/H} - 1 - 2y/H)]$ to order $J - 1$. Keep the linear Fourier phase exact. The j th profile is

$$A_{\epsilon,j}(x) = \frac{x^{-\beta}(\epsilon 2\pi i x)^j}{j! \sqrt{\pi}} \int_{\mathbb{R}} e^{-y^2 + (1-2\beta)y/H} e^{2i(T+\epsilon 2\pi x)y/H} (e^{2y/H} - 1 - 2y/H)^j dy.$$

Multiply by a smooth central cutoff. Taylor's real-phase remainder is $O_J(x^{-\Re\beta}(x/H^2)^J)$. Outside the band, nonstationary integration by parts after the u substitution bounds the kernel by $O_A(x^{-\Re\beta}(H/(T+x))^A)$, plus an exponentially small summable tail. In a bounded mollifier average the resulting absolute error is

$$O_{J,A,\epsilon} \left(T^{1+\nu+(1+\nu)\epsilon} \log T \{ (T/H^2)^J + (H/T)^A \} \right). \quad (11)$$

Indeed the central count per (h, k) is $O(Th/k)$, the outer factor is $1/h$, and $\sigma_{\alpha,-\beta}(n) \ll_{\epsilon} n^{\epsilon}$ there. The absolute h, k sum is $O(M \log M)$ after taking out T . For the tail sum integrate $(T + (k/h)n)^{-A}$, keeping its scale $Th/k \geq T/M$. Thus fixed large J, A make (11) smaller than $H/\log^B T$ for every prescribed fixed B . This uses no signed cancellation. The retained kernel is $\sum_{\epsilon=\pm 1} e(\epsilon x) \sum_{j < J} A_{\epsilon,j}(x)$. Thus each divisor sum has the additive phase $e(\epsilon np/q)$; the smooth amplitudes above do not include that arithmetic phase.

3.2 Composite transformation, with both residues

Put $h = gq, k = gp, (p, q) = 1$. The actual outer coefficient is $b_{gq} \bar{b}_{gp}/(gq)$, where $b_m = \mu(m)P(\log(M/m)/\log M)$. Squarefree support retains $(g, pq) = 1$; neither weight nor cutoff is removed. Apply the following transformation with additive numerator $A = \epsilon p$ and smooth weight $F(n) = A_{\epsilon,j}(np/q)$ for each retained profile.

For $(A, q) = 1$ define

$$D_{a,b}(s, A/q) = \sum_n \sigma_{a,b}(n) e(An/q) n^{-s}.$$

Opening both residue variables gives

$$D_{a,b}(s, A/q) = q^{-2s-a-b} \sum_{r,t=1}^q e(Art/q) \zeta(s+a, r/q) \zeta(s+b, t/q).$$

The Hurwitz Fourier formula [2] and the exact full-residue sum

$$\sum_{r,t \bmod q} e((Art + \epsilon nr + \eta mt)/q) = q e(-\epsilon \eta n m \bar{A}/q)$$

give, with $z = a + b, d = a - b$,

$$D_{a,b}(s, A/q) = 2q^{1-2s-z} (2\pi)^{2s+z-2} \Gamma(1-s-a) \Gamma(1-s-b) \cdot [\cos(\pi d/2) D_{-a,-b}(1-s, \bar{A}/q) - \cos(\pi(s+z/2)) D_{-a,-b}(1-s, -\bar{A}/q)]. \quad (12)$$

Sum r first to prove the finite identity; exactly one t survives, including nonunit n . Both Dirichlet series initially converge absolutely in the appropriate left half-plane, and continuation is meromorphic. For compact smooth F , moving the Mellin contour crosses precisely

$$q^{-1+a-b} \zeta(1-a+b) \tilde{F}(1-a) + q^{-1-a+b} \zeta(1+a-b) \tilde{F}(1-b).$$

At $a = b$ combine the poles before taking the limit: the residue is $q^{-1} \int F(x) x^{-a} (\log x + 2\gamma - 2\log q) dx$. A contour enclosing both poles is holomorphic in the shifts.

3.3 The absolute Mellin multiplier

Take $a = \alpha, b = -\beta$, now $z = \alpha - \beta, d = \alpha + \beta$, and $F(n) = A(np/q)$ for any of the finitely many profiles. Its Gaussian Fourier definition implies, for fixed $\sigma > 0$ and B ,

$$|\tilde{A}(1 - \sigma - it - z/2)| \ll_{B,\sigma} HT^{-\sigma}(1 + |t|H/T)^{-B}. \quad (13)$$

To see the uniformity, the Gaussian times $(e^{2y/H} - 1 - 2y/H)^j$ has every fixed Schwartz seminorm $O_j(H^{-2j})$. Its Fourier transform is uniformly Schwartz on the scale $(T + \epsilon 2\pi x)/H$. The x^j prefactor contributes $(T/H^2)^j \leq 1$. Every logarithmic x derivative costs at most T/H ; Mellin integration by parts gives (13). The plus-frequency primal branch is power-small with arbitrary order.

Change $u = 1 - s - z/2$ in (12). Including $2\pi/(gq)$, the oscillatory dual profile is exactly

$$\frac{4\pi}{gpq} n^{-z/2} (q/p)^{-z/2} \int_{(\sigma)} \Gamma(u - d/2) \Gamma(u + d/2) \cos(\pi u) \left(\frac{pq}{4\pi^2 n} \right)^u \tilde{A}(1 - u - z/2) \frac{du}{2\pi i}. \quad (14)$$

Start where the dual series converges absolutely. For each finite dyadic block move to any fixed $\sigma > 0$; the gamma poles lie at $\Re u \leq |\Re d|/2$. No pole is crossed for large T . The eventual summable tail bounds justify recombining all blocks. Uniform Stirling bounds give

$$|\Gamma(\sigma + it - d/2) \Gamma(\sigma + it + d/2) \cos(\pi(\sigma + it))| \ll_{\sigma} (1 + |t|)^{2\sigma-1}.$$

Consequently the absolute multiplier norm in (14) is

$$\ll HT^{-\sigma}(T/H)^{2\sigma}. \quad (15)$$

The factor $(pq/n)^{it}$ separates into three unimodular sequences. Apart from harmless small-shift powers the scale is therefore $(H/g)(pq)^{\sigma-1}(T/H^2)^{\sigma}n^{-\sigma}$. For an unbounded dual n range, charge $n^{C/\log T} \leq n^{\epsilon}$; it is not assumed bounded independently of n .

3.4 All dyadic blocks and the attributed trilinear estimate

This subsection assumes $\nu < \nu_B(\theta)$. The rational-frequency estimate below establishes the alternative ν_A range independently. On $p \asymp P, q \asymp Q, n \asymp N$ the three coefficient norms are

$$O(P^{\sigma-1/2}), \quad O(Q^{\sigma-1/2}), \quad O_{\epsilon}(N^{1/2-\sigma+\epsilon}).$$

Bettin–Chandee Theorem 1 [1], with phase parameter ± 1 , gives the product of these norms times

$$(1 + N/(PQ))^{1/2} \{ (NPQ)^{7/20+\epsilon} (P+Q)^{1/4} + (NPQ)^{3/8+\epsilon} (N(P+Q))^{1/8} \}.$$

Its arbitrary complex coefficient scope includes the separated twists; its final reciprocity argument covers both relative orders of P, Q . At $\sigma = 1/4$, the ratio to H is bounded by

$$\frac{1}{g} (T/H^2)^{1/4} \{ N^{3/5} (PQ)^{1/10} (P+Q)^{1/4} + N^{3/4} (PQ)^{1/8} (P+Q)^{1/8} \} (1 + N/(PQ))^{1/2} T^{\epsilon}.$$

Put $N_0 = TPQ/H^2$. Below N_0 , both N powers are positive and $N_0/(PQ) < 1$. Above it move to $\sigma = 1/4 + A$, gaining $(N_0/N)^A$. A fixed large A absorbs both powers, divisor losses and the

growing source parameter. If $N_0 < 1$ every integer block is in the tail and gives a stronger bound. Thus all dual blocks contribute

$$\ll \frac{H}{g} T^\epsilon \{ (T/H^2)^{17/20} (PQ)^{7/10} (P+Q)^{1/4} + (T/H^2) (PQ)^{7/8} (P+Q)^{1/8} \}.$$

Use $P, Q \leq M/g$, without assuming balance. The g sums have powers $-1 - 33/20$ and $-1 - 15/8$ and converge. Logarithmic decompositions are absorbed in ϵ , giving

$$O_\epsilon \left(HT^\epsilon \{ T^{(17/20)(1-2\theta)+(33/20)\nu} + T^{1-2\theta+(15/8)\nu} \} \right). \quad (16)$$

At $p = 1$ or $q = 1$, a convention for inversion modulo one is unnecessary: $N_0 \leq T^{1+\nu-2\theta} < 1$ in (5); a sufficiently large contour gives an arbitrarily strong saving by absolute summation. The nonoscillatory branch has $\cos(\pi d/2)$ instead of $\cos(\pi u)$. Its gamma product retains exponential vertical decay. Absolute summation on large fixed σ gives $O(H(M^2/T)^\sigma)$, negligible since $\nu < 1/2$. The small plus-frequency primal profile is handled with the same bounds and an arbitrarily large Schwartz order.

3.5 Rational-frequency averaging and a second range

We now estimate (14) without the trilinear theorem. Let $p \in [P, 2P]$, $q \in [Q, 2Q]$ and $n \in [N, 2N]$ be positive integers. Arbitrary fixed restrictions, including the original squarefree and coprime conditions, remain in the coefficient set.

Lemma 3.1 (Grouped frequency mean square, D). *For a finite sum $S(t) = \sum c_{p,q,n} e^{it \log(pq/n)}$, let $\mathcal{R} = 2N \max_{m \leq 4PQ} \tau(m)$ and $\delta = (16PQN)^{-1}$. For every $L_0 > 0$,*

$$\int_{-L_0}^{L_0} |S(t)|^2 dt \leq e\sqrt{\pi} (L_0 + 2\sqrt{\pi}/\delta) \mathcal{R} \sum |c_{p,q,n}|^2.$$

An independent bound replaces the prefactor by $2L_0 + 4\delta^{-1}(1 + \log \max\{1, J\})$, where J is the number of distinct frequencies. Both include the collision factor $\mathcal{R}$.

Proof. For distinct $x = pq/n, y = p'q'/n'$, integer cross multiplication gives $|x - y| \geq (4N^2)^{-1}$. Since $\max(x, y) \leq 4PQ/N$, their logarithms differ by at least δ . If $x = r/s$ in lowest terms, then $n = ks, pq = kr$. There are at most $2N$ possible k , and each permits at most $\max_{m \leq 4PQ} \tau(m)$ factorizations. Combining coefficients at equal frequencies into C_λ therefore gives $\sum |C_\lambda|^2 \leq \mathcal{R} \sum |c_{p,q,n}|^2$.

Exact Gaussian integration gives

$$\int_{\mathbb{R}} e^{-t^2/L_0^2} |S(t)|^2 dt = \sqrt{\pi} L_0 \sum_{\lambda, \mu} C_\lambda \overline{C_\mu} e^{-L_0^2(\lambda - \mu)^2/4}.$$

The k th ordered neighbor on either side is at distance at least $k\delta$, and $\sum_{k \geq 1} e^{-L_0^2 \delta^2 k^2/4} \leq \sqrt{\pi}/(L_0 \delta)$. Schur's row bound, or $2|C_\lambda C_\mu| \leq |C_\lambda|^2 + |C_\mu|^2$, proves the first estimate because $e^{-t^2/L_0^2} \geq e^{-1}$ on the target interval. For the second, integrate directly on that interval. Its off-diagonal Gram entries are $2 \sin(L_0(\lambda - \mu))/(\lambda - \mu)$, of magnitude at most $2/(|j - k|\delta)$. Their harmonic row sums give the stated constant. No Gaussian transform identity is used in this second derivation. $\square$

For fixed g , the exact coefficients in (14) are

$$c_{p,q,n} = b_{gq} \bar{b}_{gp} (pq)^{\sigma-1} (p/q)^{z/2} n^{-\sigma-z/2} \sigma_{-\alpha,+\beta}(n) \times \text{the actual inverse additive phase.}$$

They are independent of t . The remaining common multiplier includes $(4\pi^2)^{-it}$, the paired gamma product, the cosine and $\tilde{A}(1 - \sigma - it - z/2)$. By (13) it obeys

$$|\gamma(t)| \ll HT^{-\sigma} (1 + |t|)^{2\sigma-1} (1 + |t|/K_0)^{-B}, \quad K_0 = T/H.$$

Take $\sigma \geq 1/2$. Cauchy–Schwarz on $|t| \leq K_0$, followed by Lemma 3.1, bounds the integral by

$$\ll HT^{-\sigma} K_0^{2\sigma} (1 + PQN/K_0)^{1/2} \mathcal{R}^{1/2} \|c\|_2. \quad (17)$$

On $2^j K_0 < |t| \leq 2^{j+1} K_0$ the corresponding bound costs at most an additional $O(2^{j(2\sigma-B)})$. Choose $B > 2\sigma + 1$ and sum. This use of finite intervals is essential: a nonzero finite exponential sum has divergent whole-line unweighted square norm. The independent Gram bound has an extra logarithm; since $J \ll PQN$, it is absorbed by $T^\epsilon N^\epsilon$ in the estimates that follow.

The divisor estimate, fixed small shifts and bounded mollifier weights give

$$\|c\|_2 \ll_\epsilon (PQ)^{\sigma-1/2+\epsilon} N^{1/2-\sigma+\epsilon}, \quad \mathcal{R}^{1/2} \ll_\epsilon N^{1/2} (PQ)^\epsilon.$$

Unbounded $n^{C/\log T}$ is charged as n^ϵ . Put $a_0 = T/H^2$ and $N_0 = a_0 PQ$. At $\sigma = 1/2$, (17), including $4\pi/g$, becomes

$$\ll_\epsilon \frac{H}{g} a_0^{1/2} N^{1/2} (1 + PQN/K_0)^{1/2} (PQN)^\epsilon.$$

Expand $\sqrt{1+x} \leq 1 + \sqrt{x}$. Below N_0 the two N slopes are $1/2$ and 1 , so dyadic sums are bounded at the largest scale. Above N_0 move each finite block to $\sigma = 1/2 + D$, gaining exactly $(N_0/N)^D$. Taking $D = 4$ makes the slopes $-7/2$ and -3 before small epsilon losses. All tails converge. If $N_0 < 1$, every block lies in this tail and $N_0^D \leq N_0^{1/2}$, N_0 , so the same bound holds. No sharply supported dual cutoff is assumed.

Summing N gives

$$\ll_\epsilon \frac{H}{g} T^\epsilon \{a_0 (PQ)^{1/2} + a_0^{3/2} K_0^{-1/2} (PQ)^{3/2}\}.$$

Both p, q powers are positive. With $P, Q \leq M/g$, the gcd sums are $\sum g^{-2}$ and $\sum g^{-4}$, and outer dyadic decompositions cost only logarithms. The complete oscillatory error is consequently

$$O_\epsilon \left(HT^\epsilon \{TM/H^2 + TM^3/H^{5/2}\} \right). \quad (18)$$

Its relative exponents are $1 - 2\theta + \nu$ and $1 - (5/2)\theta + 3\nu$. Their strict negativity proves (3). The nonoscillatory branch and both residue terms are those of the exact transformation, with the absolute and coalescent bounds below; they are not omitted in this alternative estimate.

3.6 Exact Hankel symbols: an independent norm derivation

For $x \geq 1$ and d in a compact subset of $|\Re d| < 1/2$, define

$$a_\pm(x, d) = \frac{1}{\Gamma(d + 1/2)} \int_0^\infty e^{-t} t^{d-1/2} (1 \pm it/(2x))^{d-1/2} dt.$$

The U integral and its K identity [2, 13.4.4, 13.6.10], continued to $z = \mp ix$, and the connection formula 10.27.8 give exactly

$$H_d^{(1,2)}(x) = \sqrt{2/(\pi x)} e^{\pm i(x - \pi d/2 - \pi/4)} a_{\pm}(x, d).$$

For every fixed k , $x^k \partial_x^k a_{\pm} = O_k(1)$ uniformly. After differentiation the ratio $(t/x)/|1 \pm it/(2x)| \leq 2$ bounds every extra power; the remaining integrand is dominated by $e^{-t} t^{\Re d - 1/2}$ uniformly. Thus no unaccounted asymptotic remainder or singular order quotient occurs. The oscillatory Bessel combination from (14) is

$$\sin(\pi d/2) J_d(x) + \cos(\pi d/2) Y_d(x) = \sqrt{2/(\pi x)} \left[-\frac{i}{2} e^{i(x - \pi/4)} a_+ + \frac{i}{2} e^{-i(x - \pi/4)} a_- \right].$$

After $x = T/(2\pi) + Hv/(2\pi)$, the normalized profile depends only on $r = n/(pq)$. Set $\lambda = \sqrt{NT/(PQ)}$, $\mu = H\sqrt{N/(TPQ)}$. The phase $4\pi\sqrt{rx}$ has v derivative comparable to μ ; its higher derivatives divided by the first are bounded. Gaussian Schwartz bounds and integration by parts give, on fixed log- r support,

$$\|W\|_2 + (1 + \lambda)^{-1} \|W'\|_2 \ll_A (1 + \mu)^{-A}.$$

With $L_1 = 1 + \lambda$, Fourier Cauchy–Schwarz and Parseval give

$$\frac{1}{2\pi} \int |\widehat{W}(t)| dt \leq \sqrt{L_1/2} (\|W\|_2^2 + L_1^{-2} \|W'\|_2^2)^{1/2}.$$

Including the actual normalization $H/[g(PQ)^{3/4} T^{1/4}]$ and the $n^{-1/4}$ coefficient, the three norms are $O(P^{1/2}), O(Q^{1/2}), O_{\epsilon}(N^{1/4+\epsilon})$. The resulting bound before summing N is

$$\begin{aligned} & \frac{H}{g} \{ N^{17/20} (PQ)^{-3/20} (P + Q)^{1/4} + N (PQ)^{-1/8} (P + Q)^{1/8} \} \\ & \cdot (1 + N/(PQ))^{1/2} (1 + \sqrt{N/N_0})^{-A} T^{\epsilon}. \end{aligned}$$

Its dominant scale is again N_0 , reproducing (16). This route is sharper below N_0 ; both routes agree at the scale that determines the final exponents.

3.7 Residues, compact windows and general operator polynomials

The leading negative-frequency Gaussian has, after $x = T/(2\pi) + Hv/(2\pi)$, amplitude $\exp(-v^2 - i(1 - 2\beta)v/H + (1 - 2\beta)^2/(4H^2))$. Its whole- v integral is exactly $\sqrt{\pi}$. The two residue terms, including the outer factor, are consequently

$$\begin{aligned} & \frac{H\sqrt{\pi}}{g} [\zeta(1 + \alpha + \beta) q^{-1-\alpha} p^{-1-\beta} \\ & + (T/(2\pi))^{-\alpha-\beta} \zeta(1 - \alpha - \beta) q^{-1+\beta} p^{-1+\alpha}] \end{aligned}$$

up to power-small errors. Replacing the x power by its centered value costs $O(H/T)$ times logarithms; $j \geq 1$ profiles cost $(T/H^2)^j$. The absolute outer sum has only logarithmic size because its weights are $1/(gpq)$. At $\alpha + \beta = 0$, combine the residues; maximum-modulus bounds on a circle of radius $C'/\log T$ avoid a false pole loss. Fixed shift derivatives cost fixed logarithmic powers. Young's arithmetic evaluation (Lemma 2.2 and his Lemma 7) supplies the same Conrey constant

for $\nu < 1/2$. In particular the residue sum is the full diagonal main term, not an additional conjectured signed term.

For compact windows choose fixed $\eta > 0$ so that $\theta - \eta > 1/2$ and $\nu < \nu_*(\theta - \eta)$. Choose the estimate that admits the fixed ν at this narrower exponent; all its error exponents must be negative. Let $s_0 = T^{\theta-\eta}$ and $\phi_{s_0}(t) = e^{-t^2/s_0^2}/(\sqrt{\pi}s_0)$. For fixed large D set

$$w_D = \phi_{s_0} * \sum_{j < D} \frac{(-s_0^2/4)^j}{j!} w^{(2j)}.$$

The Fourier deficit $1 - e^{-u} \sum_{j < D} u^j/j!$ lies between 0 and $u^D/D!$. Thus $\|w - w_D\|_\infty \ll_D T^{-2D\eta} \log^{2D+O(1)} T$ for the stated logarithmic edge derivatives. A coarse polynomial zeta-mollifier bound on $|t| \leq 2T$ and Gaussian tails elsewhere give a power-small replacement error for sufficiently large fixed D , without using the desired moment. The signed mixture has absolute coefficient norm

$$O\left((H/s_0) \sum_{j < D} (s_0 \log T/H)^{2j}\right) = O(H/s_0).$$

Gaussian errors $O(s_0 T^{-\delta})$ therefore become $O(HT^{-\delta})$. The true loss is evaluating (16) or (18) at $\theta - \eta$; it cannot be omitted. Window mass is preserved exactly because all nonzero derivatives integrate to zero. Centers vary by $O(H)$; replacing their logarithmic normalization by $\log T$ costs only power-small errors. Applying the uniform shifted asymptotic by Cauchy to every fixed-degree Q completes Theorem 1.2 in (5). Neither a sharp indicator nor an infinite Taylor expansion has been substituted without an error estimate.

4 Distinct sign changes

4.1 An exact identity on the critical line

Let $\chi(s) = \pi^{s-1/2} \Gamma((1-s)/2)/\Gamma(s/2)$, so that $\zeta(s) = \chi(s)\zeta(1-s)$, and put $\lambda(s) = -\chi'(s)/\chi(s)$. Write $D = d/ds$ and $Q(x) = \sum_k q_k x^k$.

Lemma 4.1. *For every $k \geq 0$, $\chi(s)\zeta^{(k)}(1-s) = (-1)^k (D + \lambda)^k \zeta(s)$, where $(D + \lambda)^k$ is the k -fold composition of $f \mapsto f' + \lambda f$. Consequently $\chi(s)V(1-s) = \sum_k q_k L^{-k} (D + \lambda)^k \zeta(s)$. If $Q(x) + Q(1-x) = \beta$ identically, then*

$$E_\beta := \beta\zeta - V - \chi(s)V(1-s) = \sum_k q_k L^{-k} [(D + L)^k - (D + \lambda)^k] \zeta(s), \quad (19)$$

and with $\tilde{V} = V + E_\beta/2$ and $\omega(t) = e^{i\vartheta(t)}$,

$$\beta Z(t) = 2\Re(\omega(t)\tilde{V}(\tfrac{1}{2} + it)) \quad (t \in \mathbb{R}). \quad (20)$$

Proof. Put $f(s) = \zeta(1-s) = \chi(1-s)\zeta(s)$ and note $-\chi'(1-s)/\chi(1-s) = \lambda(s)$ because $\chi(s)\chi(1-s) = 1$. Then $f' = \chi(1-s)(D + \lambda)\zeta$ and, inductively, $f^{(k)} = \chi(1-s)(D + \lambda)^k \zeta$; since $\zeta^{(k)}(1-s) = (-1)^k f^{(k)}(s)$, the first claim follows, and the second by linearity. The polynomial identity $Q(-x) + Q(1+x) = \beta$ with $x = D/L$ gives $\beta\zeta = V + \sum_k q_k L^{-k} (D + L)^k \zeta$, which is (19). On $s = 1/2 + it$ we have $1-s = \bar{s}$, $V(\bar{s}) = \overline{V(s)}$ and $\chi(s) = \omega^{-2}$, so $\omega\chi(s)V(1-s) = \overline{\omega V(s)}$ and $\omega\zeta = Z$. Hence $\beta Z = 2\Re(\omega V) + \omega E_\beta$. The left side and $2\Re(\omega V)$ are real, so ωE_β is real, and (20) follows. $\square$

The identity (20) holds for every real Q ; the symmetry of Q makes E_β small. By Stirling's formula, $\lambda(s) = \log(|t|/2\pi) + O(1/|t|)$ and $\lambda^{(j)}(s) \ll 1/|t|$ for $j \geq 1$, uniformly in fixed vertical strips, so $\lambda - L = O(1)$ for $t \asymp T$. In normal-ordered form $(D + \lambda)^k = \sum_{m \leq k} c_{k,m} D^m$, where $c_{k,m}$ is $\binom{k}{m} \lambda^{k-m}$ plus a polynomial in $\lambda, \lambda', \dots$ of weight $k - m$ whose terms contain a derivative of λ . Hence the coefficient of D^m in $(D + L)^k - (D + \lambda)^k$ is $O(L^{k-m-1})$, and

$$E_\beta = \sum_{j < \deg Q} \varepsilon_j(s) L^{-j} \zeta^{(j)}(s), \quad \varepsilon_j(s) \ll 1/L, \quad (21)$$

uniformly for $\sigma_0 \leq \sigma \leq \sigma_1$ and $T/2 \leq t \leq 2T$. For $j \geq 1$, $L^{-j} \zeta^{(j)} = (-1)^j (V_{1+x^j} - \zeta)$, where V_{1+x^j} is the V of the polynomial $1 + x^j$, so Theorem 1.2 for the polynomials 1 and $1 + x^j$ and Cauchy-Schwarz give

$$\int w |\psi E_\beta(\sigma_0 + it)|^2 dt \ll H/L^2. \quad (22)$$

Published admissible polynomials often have $\beta \neq 1$; for instance Conrey's polynomial of degree seven has $\beta = 0.984$. For them the unnormalized difference $\zeta - V - \chi V(1 - s)$ is not small, which is why E_β is normalized by β .

4.2 The counting lemma

Lemma 4.2. *Let $\sigma_1 = \sigma_1(P, Q)$ be fixed and large, let $L = \log T_0$ be fixed with T_0 large, and let $|T - T_0| \leq 1$, $H \asymp T_0^\theta$, where T and $T + H$ are not ordinates of zeros of Z or of $\tilde{V}$ in $[1/2, \sigma_1]$. Let $N_{\tilde{V}}$ count, with multiplicity, the zeros of $\tilde{V}$ in $[1/2, \sigma_1] \times (T, T + H]$, including those on $\sigma = 1/2$. Then*

$$O(T, H) \geq N(T, H) - 2N_{\tilde{V}} - O(L).$$

Proof. Let $t_1 < \dots < t_J$ be the zeros of $\tilde{V}(1/2 + it)$ in $(T, T + H)$, of orders $k_1, \dots, k_J$, and $K = \sum k_j$. Write $\tilde{V}(s) = \prod_j (s - 1/2 - it_j)^{k_j} F(s)$ with F holomorphic and zero-free on the segment. On the line $\omega \tilde{V} = i^K p(t) \omega F$ with $p(t) = \prod_j (t - t_j)^{k_j}$ real, so by (20) $\beta Z = 2p \Re W_0$, where $W_0 = i^K \omega F$ is continuous and zero-free on $[T, T + H]$.

Let ϕ be a continuous argument of W_0 ; assume $\phi(T) < \phi(T + H)$, the other case being symmetric, and let $c_1 < \dots < c_M$ be the points of $\pi/2 + \pi\mathbb{Z}$ in $(\phi(T), \phi(T + H))$, so that $M \geq |\Delta\phi|/\pi - 1$. Put $\tau_0 = T$, $\tau_M = T + H$, and for $0 < i < M$ let τ_i be the first point with $\phi(\tau_i) = c_i + \pi/2$. Then $\tau_0 < \dots < \tau_M$ and $\cos \phi(\tau_i)$ alternates in sign, since $\phi(\tau_0) \in (c_1 - \pi, c_1)$, $\phi(\tau_M) \in (c_M, c_M + \pi)$ and $\phi(\tau_i) = c_i + \pi/2$. After moving the interior τ_i slightly, none of them is a t_j . As $\operatorname{sgn} Z(\tau_i) = \operatorname{sgn}(\beta) \operatorname{sgn} p(\tau_i) \operatorname{sgn} \cos \phi(\tau_i)$ and p changes sign only at the t_j , the function Z has opposite signs at the ends of at least $M - J$ of the disjoint intervals (τ_{i-1}, τ_i) , and each such interval contains a sign change. Hence $O(T, H) \geq |\Delta\phi|/\pi - 1 - J$.

Apply the argument principle to F on $[1/2, \sigma_1] \times [T, T + H]$. It is zero-free on the left side, and its zeros inside are the $N_{>}$ zeros of $\tilde{V}$ with $\sigma > 1/2$. On the bottom, right and top sides $\arg F = \arg \tilde{V} - \sum_j k_j \arg(s - 1/2 - it_j)$, and each factor turns by exactly $+\pi$ along these sides. Therefore

$$\Delta\phi = \Delta\vartheta + \Delta_{\text{brt}} \arg \tilde{V} - \pi K - 2\pi N_{>}.$$

The Riemann-von Mangoldt formula gives $\Delta\vartheta/\pi = N(T, H) + O(L)$. On $\sigma = \sigma_1$ we have $|V - 1| \leq 1/4$ and $|E_\beta| \ll 1/L$, so $\Re \tilde{V} > 0$ and that side contributes $O(1)$. On the horizontal sides the Backlund-Jensen argument gives $O(L)$, because $\tilde{V} = (\beta\zeta + V - \chi V(1 - s))/2 =$

$\sum_{j \leq \deg Q} a_j(s) \zeta^{(j)}(s)$ with each a_j a polynomial in L^{-1} , λ and its derivatives, so $\tilde{V}$ is holomorphic and polynomially bounded near the window and $\Re \tilde{V} \geq 1/2$ at $\sigma_1 + iT$. Since $J \leq K$ and $N_{\tilde{V}} = N_{>} + K$,

$$O(T, H) \geq N(T, H) - 2(N_{>} + K) - O(L) = N(T, H) - 2N_{\tilde{V}} - O(L). \quad \square$$

For a general window we keep $L = \log T$ and move both endpoints inward by less than one unit to avoid the finitely many exceptional ordinates, including those of zeros of $\psi \tilde{V}$; each unit interval contains $O(L)$ zeros of ζ and, by Jensen's formula, $O(L)$ zeros of $\tilde{V}$ and of $\psi \tilde{V}$ in $[\sigma_0, \sigma_1]$, so every count changes by $O(L)$. The quantity J in the proof may be replaced by the number of t_j of odd order.

A double zero of Z is not a sign change and is never counted; it is paid for in $N_{\tilde{V}}$, as an on-line zero of $\tilde{V}$ when $\deg Q = 1$ and, in the examples of Section 6, as a zero of $\tilde{V}$ to the right of the line when $\deg Q = 3$. If the on-line zeros of $\tilde{V}$ are omitted from $N_{\tilde{V}}$, the inequality fails (Section 6).

4.3 Proof of Theorem 1.3

Let $F_1 = \psi \tilde{V}$. Littlewood's lemma on $[\sigma_0, \sigma_1] \times [T, T + H]$ gives

$$2\pi \int_{\sigma_0}^{\sigma_1} n(\sigma) d\sigma = \int_T^{T+H} \log |F_1(\sigma_0 + it)| dt - \int_T^{T+H} \log |F_1(\sigma_1 + it)| dt + O(L),$$

where $n(\sigma)$ counts the zeros of F_1 in the rectangle with real part greater than σ , and $O(L)$ bounds the horizontal argument integrals: ψ and $\tilde{V}$ are polynomially bounded and have real part at least $1/2$ at $\sigma_1 + iT$ and $\sigma_1 + i(T + H)$, so the Backlund–Jensen argument bounds the variation of $\arg \psi$ and of $\arg \tilde{V}$, hence of $\arg F_1$. Each zero counted in $N_{\tilde{V}}$ has real part at least $1/2 = \sigma_0 + R/L$, and every other zero contributes nonnegatively, so the left side is at least $2\pi(R/L)N_{\tilde{V}}$.

On $\sigma = \sigma_1$, $\psi V = 1 + \sum_{n \geq 2} b_n n^{-s}$ with $|b_n| \ll d(n)(\log n)^{\deg Q}$, so its logarithm is an absolutely convergent Dirichlet series for σ_1 large and $\int \log |\psi V(\sigma_1 + it)| dt = O(1)$; moreover $|E_\beta/(2V)| \ll 1/L$ there, which contributes $O(H/L)$. On $\sigma = \sigma_0$, concavity of the logarithm, $w = 1$ on $[T, T + H]$ with $w \geq 0$, the inequality $|a + b|^2 \leq (1 + L^{-1/2})|a|^2 + (1 + L^{1/2})|b|^2$, Theorem 1.2 and (22) give

$$\int_T^{T+H} \log |F_1(\sigma_0 + it)| dt \leq \frac{H}{2} \log \left(\frac{1}{H} \int w |\psi \tilde{V}|^2 \right) \leq \frac{H}{2} \log c(P, Q, R, \nu) + O(HL^{-1/2}).$$

Hence $N_{\tilde{V}} \leq (HL/4\pi R) \log c + o(HL)$. Since $N(T, H) = (HL/2\pi)(1 + o(1))$, Lemma 4.2 gives $O/N \geq 1 - R^{-1} \log c - o(1)$. $\square$

Remark 4.3. The number σ_1 and all implied constants depend on P , Q , R and ν . For the polynomials of Section 6 the monomial coefficients of Q have size about 10^{120} and R is about 28, so the constants are very large, and for $\eta = \theta - 1/2 - \nu = 10^{-4}$ the factor $2LT^{-3\eta/4}$ of Lemma 2.1 is below one only when $\log T$ exceeds about $1.7 \cdot 10^5$. The results are purely asymptotic.

5 Transfer to simple zeros

Theorem 5.1 (D). *If $\liminf O(T, T^\theta)/N(T, T^\theta) \geq k \geq 0$, then $\liminf S(T, T^\theta)/N(T, T^\theta) \geq h(\theta; k)$.*

Proof. This is the argument of [4, proof of Theorem 1.2] with the odd-support input replaced by k . For a real even $\eta \in C_c^\infty((-\lambda/2, \lambda/2))$, $0 < \lambda < \theta$, with $\int \eta^2 = 1$, let Q_T be the pair sum of $f = \eta^2$ over the scaled zeros of the window; it is unrelated to the polynomial Q of the previous sections. With q_T, s_T, o_T the ratios of Q_T, S, O to N , the Hilbert–parity inequality [4, Proposition 2.5] reads $(q_T - s_T)(1 - o_T) \geq 2(1 - s_T)^2$, and Wang’s theorem [14, Theorem 2.2] gives $q_T \rightarrow \mathcal{C}(f)$. If $O = N$, every zero of the window is simple and critical and $s_T = 1$. Otherwise $1 - o_T > 0$ and the inequality gives $q_T \geq s_T$; for every $\epsilon > 0$ and all large T , $q_T \leq \mathcal{C}(f) + \epsilon$ and $o_T \geq k - \epsilon$, and enlarging the nonnegative factors gives $2(1 - s_T)^2 \leq (\mathcal{C}(f) + \epsilon - s_T)(1 - k + \epsilon)$. Hence $\liminf s_T \geq R(\mathcal{C}(f), 1 - k)$ with $R(A, b) = (4 - b - \sqrt{b(b + 8(A - 1))})/4$, the smaller root. Letting f tend to the cosine density at fixed λ , so that $\mathcal{C}(f) \rightarrow 2 - c(\lambda)$, and then $\lambda \rightarrow \theta$, gives $R(2 - c(\theta), 1 - k) = h(\theta; k)$. $\square$

For fixed $k < 1$, $h(\theta; k)$ increases with c and with k , since $\partial_k[(1 - k)(9 - k - 8c)] = -(10 - 2k - 8c) < 0$ for $c < 1$. As $c'(\theta) = \frac{1}{2} \cot^2(\theta/\sqrt{2}) > 0$, if $h(\theta_*, \kappa) > 0$ for a detector satisfying (5) at θ_* , then the same detector is admissible and $h(\theta; \kappa) > 0$ for every $\theta \in [\theta_*, 1)$. The linear branch $(c + 2k)/3$ in (6) is the transfer of [4, Theorem 1.4] with the same input.

6 Certified detector constants

We use Q in the Chebyshev family

$$Q(y) = 1 - y + \sum_{j=1}^{100} x_j (T_{2j+1}(1 - 2y) - (1 - 2y)),$$

which has degree 201, $Q(0) = 1$, $Q(1) = 0$ and $Q(y) + Q(1 - y) = 1$ for all rational x_j , because T_{2j+1} is odd with $T_{2j+1}(1) = 1$, and $P(x) = S(rx)/S(r)$ with S the Taylor polynomial of $\sinh$ of degree 13. The coefficients x_j are rationals with denominator 10^{40} obtained by minimizing c numerically, and R, r, ν are rational. With $B_P = \int_0^1 P^2$, $C_P = \int_0^1 P'^2$ and $\int_0^1 PP' = 1/2$, and since $\int_0^1 w_R w'_R = (e^{2R}Q(1)^2 - Q(0)^2)/2 = -1/2$,

$$c(P, Q, R, \nu) = \frac{1}{2} + \frac{C_P}{\nu} \int_0^1 e^{2Rv} Q(v)^2 dv + \nu B_P \int_0^1 e^{2Rv} (RQ(v) + Q'(v))^2 dv.$$

Each integral $\int_0^1 e^{2Rv} F(v) dv$ of a rational polynomial equals $e^{2R}a(F) + b(F)$ with rational a, b , by the recursion $\int_0^1 e^{tv} v^k dv = e^t a_k + b_k$, $a_k = 1/t - (k/t)a_{k-1}$, $b_k = -(k/t)b_{k-1}$, $a_0 = 1/t$, $b_0 = -1/t$. Hence $c = Ae^{2R} + B$ with exact rationals A, B , and $\kappa = 1 - R^{-1} \log c$ needs one exponential and one logarithm in ball arithmetic (Arb, 4000 bits). The admissibility identities are verified in exact rational arithmetic.

In our numerical optimization the ratio κ/ν did not exceed about 0.7173 for this mollifier shape at small ν ; the polynomials of Conrey, Farmer, Kwan, Lin and Turnage-Butterbaugh [5] give about 0.682ν . Linear Q does not suffice at these lengths: $Q = 1 - x$, $P = x$ needs $\nu > 0.1928$ for $\kappa > 0$. The same code reproduces Young’s $c = 2.35006777\dots$ for $P = x$, $Q = 1 - x$, $R = 1.3$, $\nu = 1/2$, and Conrey’s $\kappa > 0.4088$ for his polynomial of degree seven with $\nu = 4/7$.

An independent program evaluates Q by the Chebyshev recurrence directly from the x_j and P by its series, separates the double integral by Fubini into one-dimensional integrals, computes each by validated Arb quadrature, and recomputes $c(\theta)$ and h with 100-digit interval arithmetic in `mpmath`. Every enclosure overlaps the primary one, and all thresholds hold again. With $\nu = \theta - 1/2 - 10^{-4}$, the directed values of $h(\theta; \kappa)$ at $\theta = 0.534, 0.535, 0.54, 0.5459, 0.55$ exceed

Table 1: Certified detector constants. Lower bounds are rounded down.

ν	R	r	$c(P, Q, R, \nu)$	$\kappa >$	$\kappa/\nu >$
0.0199	48.391	0.7702	$5.1998 \cdot 10^{20}$	0.0142729911	0.717235
0.0299	32.244	0.7708	$5.0471 \cdot 10^{13}$	0.0214483010	0.717334
0.0339	28.439	0.7708	$1.1235 \cdot 10^{12}$	0.0243176419	0.717334
0.0349	27.625	0.7709	$4.9777 \cdot 10^{11}$	0.0250349766	0.717334
0.0399	24.163	0.7709	$1.5614 \cdot 10^{10}$	0.0286216496	0.717334
0.0458	21.05	0.7708	$6.9431 \cdot 10^8$	0.0328539236	0.717334
0.0499	19.321	0.7709	$1.2321 \cdot 10^8$	0.0357949954	0.717334
0.0999	9.648	0.7707	$7.7588 \cdot 10^3$	0.0716640051	0.717357

$1.4806994 \cdot 10^{-5}$, 0.0015246940 , 0.0090231376 , 0.0177638490 and 0.0237708528 ; at $\theta = 0.5459$ the ratio to the rank-six Selberg value $1.7764518 \cdot 10^{-5}$ exceeds 999.96.

Numerical controls support Lemma 4.2. At $T = 10^4, 10^5, 10^6$ and for Q of degrees 1, 3, 5, 7, including Conrey's polynomial and polynomials with $\beta \neq 1$, the direct evaluation of $\chi(s)V(1-s)$ agrees with Lemma 4.1, ωE_β is real and (20) holds to the working precision. On windows at these heights the sign changes of Z , the level crossings of $\arg W_0$ and the zero counts coincide. For the test functions $f = \zeta \prod_j ((s-1/2)^2 + \gamma_j^2)^{e_j}$ with planted double ($e_j = 1$) and triple ($e_j = 2$) critical zeros, the inequality of Lemma 4.2 holds with equality in the double-zero cases, double zeros are not counted, and the variant that omits the on-line zeros of $\tilde{V}$ fails for $\deg Q = 1$.

7 Scope

The rational-frequency estimate uses elementary finite mean-square bounds, classical special-function identities and Young's arithmetic evaluation. The complementary range uses the attributed Bettin–Chandee theorem. The counting proof is localized with full weight for detector zeros on the line. The simple-critical consequence additionally uses Wang's recent short-interval pair theorem and the finite parity inequality. The complete estimates retain collisions, tails and the compact smoothing margin. Exact and interval controls verify finite normalizations and constants; they do not substitute for the universal analytical proof, establish external peer review or prove worldwide novelty. All statements are asymptotic and give no effective height. Very large fixed detector coefficients and smoothing orders can make constants astronomical. The range does not reach positivity for every $\theta > 1/2$, and no universal simplicity statement or RH proof follows.

At $\theta = 527/1000$, $\nu = 499/10000$, choose $\eta = 1/100000$. The narrower Gaussian exponent is $52699/100000$; the two error exponents are exactly $-51/12500$ and $-6711/40000$. The two independent arithmetic paths give $h > 0.0005947542001$. An exact rational reconstruction of the raw multiplier, coefficient, collision, spacing and gcd exponents checks 6,426 configurations, including unbalanced and far-tail blocks. The universal slope argument supplies summability beyond those samples. 432 frequency blocks with 524,400 triples independently verify rational grouping, spacing and collision inequalities. A coprime ratio-one group of six triples has normalized interval mass 36 rather than the incorrect diagonal-only value 6. A nonzero constant exponential has divergent whole-line unweighted square norm; the proof uses only finite intervals and summable weighted bands. Excessive narrower-window loss, zero strict margin and an uncontrolled dual tail fail exact controls.

Data availability. The programs, exact parameters, outputs and independent replays are

archived as computations EXP-010 and EXP-024–029 in the repository [9, 10, 11].

Declaration of generative AI use. Generative AI tools were used for source research, derivation, review and implementation. Two analytical derivations and the stated numerical controls were used to test the claims. This preprint has not undergone external peer review.

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